PROGRAM – 6

Date:03/06/2025

Objective: To implement multithreading in Java.

```
class Main{
   public static void main(String a[]){
      Customer c= new Customer();
      Thread t1=new Thread(){
      public void run(){
         c.withdraw(2000);
         System.out.println("After withdrawal, amount is "+c.amount);
         }
      };
      Thread t2=new Thread(){
      public void run(){
         c.deposit(5000);
         System.out.println("After deposit, amount is "+c.amount);
         }
      };
      t1.start();
      t2.start();
   }
}
class Customer{
   int amount = 0;
   int flag = 0;
   synchronized void withdraw (int amount){
      System.out.println(Thread.currentThread().getName() +": is going to withdraw.");
      if (flag==0){
         try {
            System.out.println("Waiting...");
            wait();
         }
         catch (Exception e){
            System.out.println("Something went wrong");}
      }
      this.amount-=amount;
      System.out.println("Withdrawal compeleted!");}
   synchronized void deposit(int amount){
      System.out.println(Thread.currentThread().getName()+" : is going to deposit.");
      this.amount+=amount;
      notifyAll();
      System.out.println("Deposit compeleted!");
      flag=1;
   }
}
```

OUTPUT

```
C:\Users\ASUS\OneDrive\Desktop\java>javac MainTd.java

C:\Users\ASUS\OneDrive\Desktop\java>java MainTd
Thread-1 : is going to deposit.
Deposit compeleted!
Thread-0: is going to withdraw.
Withdrawal compeleted!
After deposit, amount is 5000
After withdrawal, amount is 3000

C:\Users\ASUS\OneDrive\Desktop\java>
```